

The Seizure Prognosis of Juvenile Myoclonic Epilepsy

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Abstract: Thirty-two patients with juvenile myoclonic epilepsy (JME) were studied to evaluate the seizure prognosis. The response to antiepileptic drugs was excellent in 68%, but the patients, who had much more focal discharges on EEG and were sensitive to neuropsychological EEG activations at the beginning of treatment, had an unfavorable outcome. A combination of absence seizure alone resulted in the excellent prognosis for both absence and myoclonic seizures, and a combination of generalized tonic-clonic seizure on awakening related to rare myoclonic seizures. These findings suggest that the outcome of JME would be predicted by the EEG abnormality and the combination of the other types of seizures, which are probably determined by the pathophysiology at the beginning of treatment.

Key Words: *prognosis, juvenile myoclonic epilepsy, myoclonic seizure, neuropsychological EEG activation, grand mal epilepsy on awakening*

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INTRODUCTION

Among epilepsies and epileptic syndromes, juvenile myoclonic epilepsy (JME) is classified into a specific type of age-related idiopathic generalized epilepsy. JME is closely related to juvenile absence epilepsy and grand mal epilepsy on awakening in terms of the clinical features and is characterized by some pathophysiological aspects: 1) a high incidence of paroxysmal discharges by intermittent photic stimulation (called "photosensitivity" here, irrespective of seizure precipitation)⁷ and 2) provoking effects of sleep-wake cycle ("seizure on awakening")⁶ and mental activities^{1,2} on seizure occurrence.

Present antiepileptic drugs, especially sodium valproate, are very effective to control the seizures in JME, while the discontinuation or reduction of such drugs often results in a relapse, even if the seizures would be well controlled over a long period.⁴

In this paper, the author will discuss the factors related to the long-term prognosis of the seizures in JME.

SUBJECTS AND METHODS

Thirty-two patients with JME, who had myoclonic seizures beginning after the age of 8 for at least 3 years and had no neurological or mental deficits, were selected for this investigation. The EEG examinations, including a neuropsychological activation³ and a photic stimulation with geometric patterns and colors⁵ as a special EEG ac-

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tivation procedure, were repeatedly carried out from the beginning of treatment to the present investigation. Simultaneous video-EEG and EMG (upper extremities) monitoring during the neuropsychological EEG activation, which can be very potent to provoke seizures in laboratory¹, was added to detect a subtle myoclonic seizure in the recent EEG examinations. All of the patients were under medication of sodium valproate with or without other anti-epileptic drugs. The author asked all of the patients about the clinical history and present status in detail.

RESULTS

Clinical Course of Duration

According to the seizure course of duration, the patients were divided into 3 groups; 1) short course group suffering from 3 to 9 years (N=8, mean age=19 years old), 2) intermediate group suffering from 10 to 19 years (N=15, mean age=28 years old), and 3) long course group suffering from 20 to 39 years (N=9, mean age=42 years old). No differences among the 3 groups were found in the age of seizure onset, sex distribution, family history of epilepsy, febrile convulsion or previous history related to epilepsy. The rate of incomplete seizure control by antiepileptic drugs and the incidence of combination of generalized tonic-clonic seizures were slightly higher in the long course group than in the other groups. The long course group had much more focal discharges (central- or frontal-dominant bilateral spike-wave discharges^{1,2}) on EEG both at the beginning of treatment and this investigation. The effect of sleep activation on EEG was almost similar for the 3 groups, but for the long course group photic stimulation was less effective and neuropsychological activation was more effective both at the beginning of treatment and this investigation compared with the other groups.

Frequency of Myoclonic Seizure

According to the frequency of myoclonic seizures under medication at this investigation, the patients were classified into 3 groups; 1) frequent group in which seizures occur every week or every month (N=10, mean age=31 years old), 2) rare group in which seizures incidentally occur (N=12, mean age=29 years old), and 3) remission group in which seizures disappear for at least 3 years (N=10, mean age=31 years old). Note that most patients in the rare group had not realized their fits until the author pointed out the subtle myoclonic seizures detected by the simultaneous video-EEG and EMG monitoring.

No differences among the 3 groups were found in the age of seizure onset, clinical course of duration, sex distribution, previous history related to epilepsy, combination of seizures, control of generalized tonic-clonic seizure, or incidences of generalized and focal discharges on EEG. In the frequent group, sleep and neuropsychological activations except for photic stimulation were much more effective than in the other groups at the beginning of treatment and this investigation.

Generalized Tonic-Clonic Seizure (GTCS)

Twenty-four (75%) out of the 32 patients had GTCS; 8 patients with rare GTCSs, 1 patient with GTCSs only in sleep, 4 patients with GTCSs irrespective of the sleep-wake cycle, and 11 patients with GTCSs on awakening. Eight patients without GTCS were younger in the seizure onset than those with GTCS. Eleven patients with GTCS on awakening were classified both into JME and grand mal epilepsy on awakening and showed the characteristics as follows. They had high incidences of febrile convulsion, trivial head injury which unusually results in epilepsy, and paroxysmal discharges on an awake EEG. Neuropsychological EEG activation was less effective in these patients. Most of

them were classified into rare and remission groups in terms of frequency of myoclonic seizure, although GTCSs were controlled for 3 years in only 55% of them.

Absence Seizure

Nineteen (56%) out of the 32 patients had absence seizures. Since their absence seizures were rare in frequency (non-pyknolepsy) and short in duration, it was precisely difficult to evaluate the relationship between the absence seizure and prognosis of JME. However, the seizures were well controlled for 3 years in the patients with absence seizure alone (100%) compared with those with GTCS (55%) and those with GTCS and absence seizures (31%).

DISCUSSION

In general, the myoclonic seizures in JME may begin at around adolescence (11–19 years old in 91% of our patients) and then gradually improve with much inter-individual differences. Our patients in the long course group were unfavorably controlled by antiepileptic drugs, and they showed more focal discharges on EEG, less photosensitivity and much positive effect of neuropsychological EEG activation on paroxysmal discharges even at the beginning of treatment. Therefore, it is likely that the severity of disease process itself rather than the psychosocial factors might be crucial to determine the prognosis of JME.

Myoclonic seizures were well controlled by antiepileptic drugs in 68% of the patients (rare group and remission group). Our patients in the frequent group showed a high incidence of family history of epilepsy and much positive effects of sleep and neuropsychological activations on paroxysmal discharges at the beginning of treatment, suggesting that responsiveness to drugs or frequency of myoclonic seizure might be regulated by the disease process

and predicted by the EEG activation procedure.

A combination of absence seizure and/or GTCS would influence the prognosis of JME as follows: (1) no correlation was found between the frequency of myoclonic seizure and drug control of GTCS; (2) the patients with GTCS on awakening showed some clinical and EEG characteristics and had rare myoclonic seizures; (3) the patients with a combination of absence seizure alone had excellent prognosis in comparison with those with GTCS. These facts may lead to some assumptions that the JME would be heterogeneous, that GTCS and/or absence seizure would alter the prognosis of JME, or that each seizure type would have the different prognosis. A further study will be required to explain the facts.

Lastly the author emphasized that it would be useful to predict the prognosis in terms of the pathophysiology assessed by the special EEG activation procedures described here.

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